

Task 1:

Key Differences Between IPv4 and IPv6 Addressing

Objective: Compare the primary architectural, formatting, and operational differences between IPv4 and IPv6 protocols in a structured comparison table.

1. IPv4 vs. IPv6 Comparison Table:

Comparison Parameter	IPv4 (Internet Protocol Version 4)	IPv6 (Internet Protocol Version 6)
Address Bit Length	32-bit binary address space.	128-bit binary address space.
Address Format & Representation	Represented in Dotted-Decimal notation	Represented in Hexadecimal notation separated by colons (e.g., 2001:0db8:85a3::8a2e:0370:7334).

Comparison Parameter	IPv4 (Internet Protocol Version 4)	IPv6 (Internet Protocol Version 6)
	(e.g., 192.168.1.1).	
Total Address Space Capacity	Supports approximately 4.3 billion (2^{32}) unique IP addresses.	Supports over 340 undecillion (2^{128}) unique IP addresses.
Address Allocation & NAT Requirement	Relies heavily on NAT (Network Address Translation) due to severe IPv4	Direct end-to-end global connectivity with No NAT required due to vast address space.

Comparison Parameter	IPv4 (Internet Protocol Version 4)	IPv6 (Internet Protocol Version 6)
	address exhaustio n.	
Configurati on Mechanism	Configure d manually or automatic ally using a DHCP Server .	Supports SLAAC (Stateless Address Autoconfiguration) as well as DHCPv6.
Header Architectur e	Variable length header (20–60 bytes) with optional fields, requiring	Simplified, fixed-size header (40 bytes) with no checksum field for faster router processing.

Comparison Parameter	IPv4 (Internet Protocol Version 4)	IPv6 (Internet Protocol Version 6)
	checksum calculatio n at each router hop.	

Task 2:

Static IPv4 Address Allocation for Home Wi-Fi Devices

Objective: Design a static IPv4 addressing scheme for 4 home devices (Laptop, Smartphone, Smart TV, and Network Printer) on the 192.168.10.0/24 local network.

1. Network Configuration Parameters:

- **Network Address:** 192.168.10.0
 - **CIDR Prefix:** /24
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway (Wi-Fi Router IP):** 192.168.10.1
 - **Broadcast Address:** 192.168.10.255
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2. Device Static IP Assignment Table:

Device Name	Assigned Static IPv4 Address	Subnet Mask	Default Gateway	Device Function / Description
Wi-Fi Router	192.168.10.1	255.255.255.0	N/A (Gateway Node)	Default Gateway for Internet routing.
Laptop	192.168.10.10	255.255.255.0	192.168.10.1	Primary workstation for work & development.
Smartphone	192.168.10.11	255.255.255.0	192.168.10.1	Mobile device on home wireless network.

Device Name	Assigned Static IPv4 Address	Subnet Mask	Default Gateway	Device Function / Description
Smart TV	192.168.10.20	255.255.255.0	192.168.10.1	Streaming media device.
Network Printer	192.168.10.50	255.255.255.0	192.168.10.1	Shared network resource (Static IP avoids IP changes).

3. Addressing Design Rationale:

- 1. Subnet Mask Uniformity:** All devices use 255.255.255.0 because /24 reserves 24 bits for the

network portion (192.168.10) and 8 bits for host addresses.

2. **Gateway Consistency:** All end-user devices share 192.168.10.1 as the Default Gateway to forward external internet traffic.
3. **Collision Avoidance:** Static IP assignments ensure no two devices share an address, preventing IP conflict errors on the LAN.

Task 3:

Subnetting Calculation for 192.168.1.0/27 Network

Objective: Calculate the total valid subnets, usable hosts per subnet, block size, and detail the address ranges for the first three subnets on a 192.168.1.0/27 network.

1. Subnetting Calculations Summary:

- **Base Network:** 192.168.1.0
 - **CIDR Prefix:** /27
 - **Subnet Mask:** 255.255.255.224
 - **Borrowed Subnet Bits (n):** $27 - 24 = 3$ bits
 - **Number of Valid Subnets:** $2^3 = 8$ Subnets
 - **Remaining Host Bits (h):** $32 - 27 = 5$ bits
 - **Usable Hosts per Subnet:** $2^5 - 2 = 32 - 2 = 30$ Usable Hosts/Subnet
 - **Block Size / Increment:** $256 - 224 = 32$
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2. First 3 Valid Subnet Address Ranges Table:

Sub net No.	Netwo rk Addres s	First Usable Host IP	Last Usable Host IP	Broad cast Addres s	Full Addres s Range
Sub net 1	192.16 8.1.0	192.16 8.1.1	192.16 8.1.30	192.16 8.1.31	192.16 8.1.0 – 192.16 8.1.31
Sub net 2	192.16 8.1.32	192.16 8.1.33	192.16 8.1.62	192.16 8.1.63	192.16 8.1.32 – 192.16 8.1.63
Sub net 3	192.16 8.1.64	192.16 8.1.65	192.16 8.1.94	192.16 8.1.95	192.16 8.1.64 – 192.16 8.1.95

Task 4:
Bitwise AND Calculation and Subnet Analysis for
10.0.5.17/24

Objective: Calculate the network address, broadcast address, and valid host IP range for the IP 10.0.5.17 with subnet mask 255.255.255.0 using bitwise AND operations.

1. Subnet Mask & CIDR Conversion:

- **Given IP Address:** 10.0.5.17
 - **Given Subnet Mask:** 255.255.255.0
 - **Binary Subnet Mask:**
11111111.11111111.11111111.00000000
 - **CIDR Prefix Length: /24** (24 contiguous network bits)
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2. Bitwise AND Network Address Calculation:

IP Address (10.0.5.17) : 00001010 . 00000000 . 00000101
. 00010001

Subnet Mask (255.255.255.0): 11111111 . 11111111 .
11111111 . 00000000

Bitwise AND (Network ID) : 00001010 . 00000000 .
00000101 . 00000000 => 10.0.5.0

3. Subnet Parameter Results Summary:

Subnet Parameter	Calculated Value	Parameter Description
CIDR Notation	10.0.5.17/24	IP with 24 network bits and 8 host bits.
Network Address	10.0.5.0	Base identifier for the subnet.
First Usable Host IP	10.0.5.1	Network Address + 1
Last Usable Host IP	10.0.5.254	Broadcast Address - 1
Broadcast Address	10.0.5.255	Network address with all host bits set to 1.
Valid Host IP Range	10.0.5.1 to 10.0.5.254	Total of 254 usable host addresses ($2^8 - 2$).

Task 5:

Online Subnet Calculator Verification & Result Mapping

Objective: Verify the manually calculated subnet parameters (Network Address, Broadcast Address, Host Range) for 10.0.5.17/24 using an online Subnet Calculator tool.

1. Online Subnet Calculator Input & Output Parameters:

- **Input Parameters Entered:**
 - **Network IP Address:** 10.0.5.17
 - **Subnet Mask:** 255.255.255.0 (/24)
 - **Tool Output Results Verified:**
 - **Network Address:** 10.0.5.0
 - **Broadcast Address:** 10.0.5.255
 - **Usable Host IP Range:** 10.0.5.1 – 10.0.5.254
 - **Total Usable Hosts:** 254
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2. Screenshot Placeholder & Output Verification Statement:

[Insert Screenshot of Subnet Calculator Output for 10.0.5.17/24 Here]

- **Verification Statement:**

"The manual bitwise calculations performed in Task 4 perfectly matched the output generated by the online subnet calculator tool, confirming that the Network Address is 10.0.5.0, the Broadcast Address is 10.0.5.255, and the usable IP range is 10.0.5.1 to 10.0.5.254."